

Diagnose before you regularize

Fit on training data, make choices on validation data, and open the test set once.

THE QUESTION

A model can drive training error to zero. How do we make it do well on data it has never seen?

THE INTUITION

Where the training data are silent, many fits look equally good. A regularizer states a preference at one site, and validation decides how strong that preference should be.

KEY TAKEAWAY

Diagnose with learning curves, change one site at a time, let validation set the knob, and open the test set once.

plot curves → **diagnose** → pick one site → sweep knob on val → restore best checkpoint → test once

1 Read both curves first: level, gap, trend

Pattern

val $\gg$ train
both high
val turns up
poor, small gap

Diagnosis

high variance
high bias
overtraining
underfit

First move

regularize, or add valid data
better model or optimization
restore the best checkpoint
do not regularize harder

TRAIN

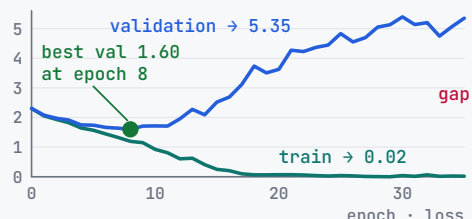
fits the parameters θ

VALIDATE

chooses method, strength, schedule, checkpoint

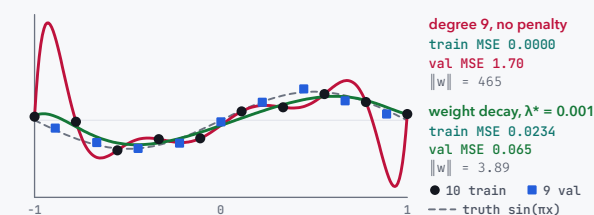
TEST

estimates performance once, after choices are frozen



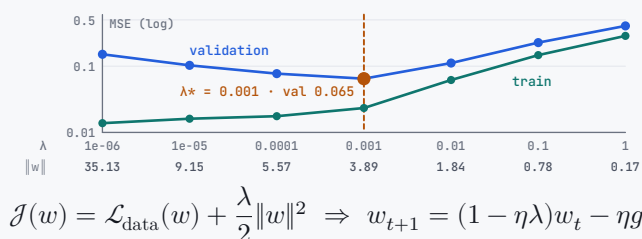
Measured CNN on 2,000 CIFAR-10 images: at epoch 35, train accuracy 99.6% vs validation 44.2%.

2 A perfect fit can fail



Ten noisy points from $\sin(\pi x)$, $\sigma = 0.25$; features $1, x, \dots, x^9$. Ten weights through ten points: **zero training error**, validation MSE 1.70 (27 $\times$ the noise floor). Training loss cannot tell a good fit from memorization.

5 Weight decay: shrink, then step

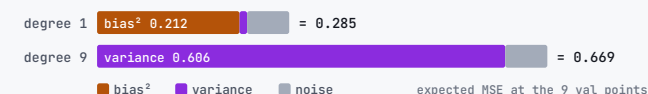


$$\mathcal{J}(w) = \mathcal{L}_{\text{data}}(w) + \frac{\lambda}{2} \|w\|^2 \Rightarrow w_{t+1} = (1 - \eta\lambda)w_t - \eta g_t$$

With $w = 5$, $g = 3$, $\eta = 0.1$, $\lambda = 0.2$: shrink $\times 0.98 \rightarrow 4.90$, then step $-0.30 \rightarrow 4.60$. Sweep λ on a log scale: too little leaves high variance, too much underfits.

With adaptive optimizers, AdamW decouples the shrinkage from the adaptive gradient scaling.

3 The bias-variance lens

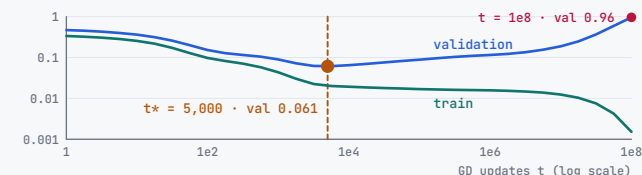


expected squared error = $\text{bias}^2 + \text{variance} + \text{noise}$
with $\text{bias} = \mathbb{E}[\hat{f}(x)] - f(x)$ and $\text{variance} = \text{Var}(\hat{f}(x))$ over redraws of the noise. Degree 1 is too rigid; degree 9 chases noise. Regularization accepts a little more bias to cut variance; validation tests whether the trade pays off.

4 Compare at fixed conditions

Fix the split, model family, optimizer budget, augmentation policy and evaluation; change one knob. Training loss may rise: ask whether validation performance and stability improve.

6 Early stopping keeps the best checkpoint



$t^* = \arg \min_t \mathcal{L}_{\text{val}}(w_t)$. Plain GD on the same degree-9 model, no penalty: validation bottoms out at $t^* = 5000$, then the fit chases noise.

- 1 save when validation improves
- 2 stop after the patience is used up
- 3 restore the saved best state

The last epoch is not automatically the selected model.

Operate each regularizer correctly

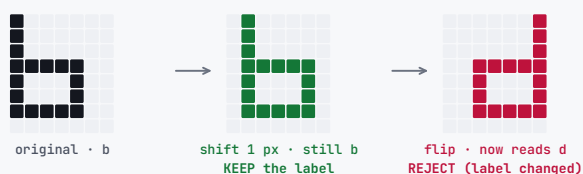
Know what changes during training, what stays fixed, and which knob validation chooses.

7 Choose the intervention site

Site	Method, knob	Preference
objective	weight decay, λ	smaller weights
loop	early stopping, t^*	shorter trajectory
data	augmentation, s	valid nearby views
model	dropout, p_{drop}	redundant features
targets	label smoothing, ε	finite confidence

s = augmentation strength. Mixup and CutMix act on examples and targets. Change one site at a time when possible.

8 Augmentation: valid variation only

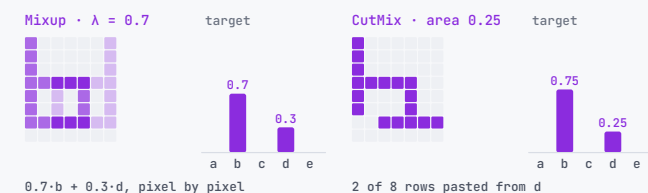


$A_\tau(x, y) = (g_\tau(x), h_\tau(y))$. For every transform decide:

- KEEP** the label is unchanged
- UPDATE** move boxes, masks, keypoints, spans, times
- REJECT** the example is no longer task-valid

Train transforms are random; validation and test transforms are deterministic. Inspect real batches before training.

9 Mixup and CutMix: mix inputs and targets



$$\lambda \sim \text{Beta}(\alpha, \alpha), \quad \tilde{x} = \lambda x_i + (1 - \lambda)x_j, \\ \tilde{y} = \lambda y_i + (1 - \lambda)y_j$$

The model is asked to vary smoothly between examples. CutMix weights the target by the pasted area fraction. Here $\text{CE} = -0.7 \ln p_b - 0.3 \ln p_d$ is smallest (0.611) at $p = \tilde{y}$; a confident $p_b = 0.99$ costs 1.804. Validate α , and ask whether the mixed input and the soft target make sense for the task.

13 A step-by-step recipe, then a final audit

- 1 fit a reproducible baseline; plot learning curves
- 2 add only task-valid augmentation
- 3 sweep weight decay on a log scale
- 4 add one targeted method if curves justify it
- 5 freeze recipe and checkpoint; test once

10 Dropout: practice with missing features

h	2	4	6	8	$q = 1 - p_{\text{drop}} = 0.5$
mask A	1	0	1	0	train: survivors $\times 1/q$
train A	4	0	12	0	eval: no mask, no scaling
train B	0	8	0	16	$E[\text{train } h_3] = 0.5 \cdot 12 + 0.5 \cdot 0 = 6 = h_3$
eval	2	4	6	8	mean of 2,000 masks: (1.97, 4.01, 6.05, 8.19)

illustrative activations

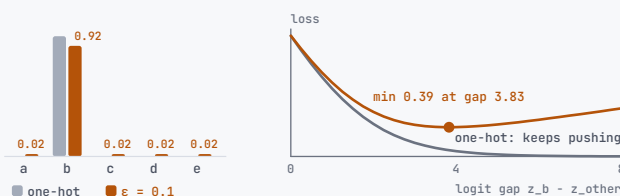
With keep probability $q = 1 - p_{\text{drop}}$:

$$m_i \sim \text{Bernoulli}(q), \quad \tilde{h}_i = \frac{m_i h_i}{q}, \quad \mathbb{E}[\tilde{h}_i] = h_i.$$

- TRAIN** fresh mask each pass; survivors scaled by $1/q$
- EVAL** all units active; no mask, no extra scaling

PyTorch `nn.Dropout(p=0.5)`: this p is the drop probability. `model.eval()` switches dropout off; `inference_mode()` changes autograd. They solve different problems.

11 Label smoothing: a finite target



$$\tilde{y} = (1 - \varepsilon)y + \frac{\varepsilon}{K}\mathbf{1}, \quad \frac{\partial \mathcal{L}}{\partial z} = p - \tilde{y}.$$

With $K = 5, \varepsilon = 0.1$: correct class 0.92, each wrong class 0.02. The loss is minimized at a finite logit gap, $\ln(0.92/0.02) \approx 3.83$, so the gradient stops demanding infinite separation.

Validate ε : too much smoothing can erase useful confidence or class-similarity structure.

12 Measured: one change at a time



Same CNN, split and budget; one regularizer per run. Crops and flips helped most. One seed only: a teaching case rather than a benchmark.

Weights constrain parameters · stopping constrains time · augmentation adds valid views · dropout disrupts pathways · smoothing softens targets

Polynomial numbers: the Lecture 10 deck's seeded example. CIFAR-10: measured (2,000 train images, 666,538-parameter CNN, 35 epochs, one seed). Dropout activations are illustrative. Slides, notebooks and summary videos: nipunbatra.github.io/dl-2026.